



HARISHKUMARAN V

EMERGING B.TECH GRADUATE

CONTACT

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SKILLS

- Project Management
- Teamwork
- Time Management
- Leadership
- Effective Communication
- Critical Thinking

TECHNICAL SKILLS

- Proficient in ESP-IDF
- Peripheral interfacing
- Data logging and visualization
- Digital Circuit Design
- Signal Filtering and Processing

ACHIEVEMENT

- ACM SIGBED - Treasurer
- DIGICON - Event coordinator
- CUBE XTREME - 1st Place
- EDUAITHON - Winner

LANGUAGES

- English (Fluent)
- Tamil (Fluent)
- Mandarin (Basic)
- Telugu (Basic)

ABOUT ME

I am an emerging third-year B.Tech student specializing in **Electronics and Communication Engineering (ECE)**, committed to building a strong foundation in both **theoretical and practical aspects** of my field. With a dedicated approach to my studies, I aim to deepen my understanding of key concepts such as **circuit design, signal processing, and communication systems**.

EDUCATION

Kalasalingam Academy of Research Education., Virudhunagar	2023-2027
Electronics and Communication Engineering (ECE)	
Embedded IoT	
9.36 CGPA*	
Kalasalingam Academy of Research Education., Virudhunagar	2023-2027
Aeronautical engineering	
Minors in Unmanned Aerial Vehicle (UAV)	
8.6 CGPA*	
Maharishi Vidya Mandir School., Hosur	2022-2023
Senior Secondary Certificate	
8.1 CGPA	
The Titan School., Hosur	2019-2022
Higher Secondary Certificate	
8.1 CGPA	

WORK EXPERIENCE

Intern - Watch Division, Titan Company Limited
Duration: May 12 - June 12, 2025
Location: Tool Room, Maintenance, and R&D Departments

- Gained hands-on exposure to electrical systems of advanced machines such as ESSA 15T, Bruderer BSTA 200-60, and MIKRON HEM 1000.
- Troubleshoot and maintained high-speed servo motors and control circuits.
- Designed and implemented automation systems using PLCs and image processing.

PROJECTS

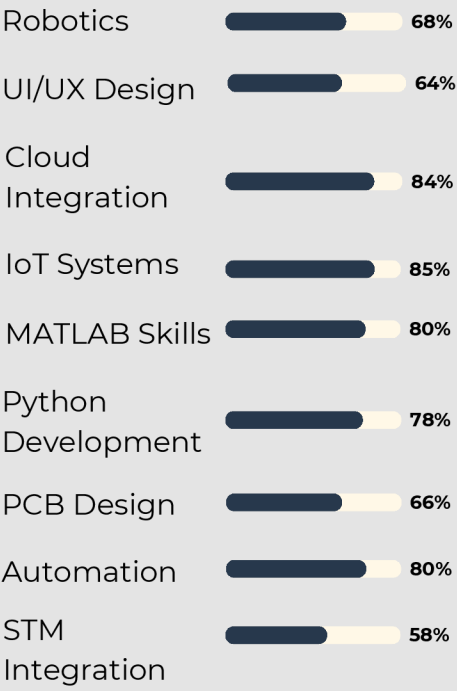
Data-Driven Interactive Smith Chart Visualization for Transmission Lines

Duration: Jan 2026
Role: Technical Developer
Tools/Technologies: HTML, CSS, JavaScript, Interactive Simulation, RF Concepts (Impedance, VSWR, Smith Chart), Data Visualization

IoT Monitoring & Digital Ledger for Pharmaceutical Supply Chain (PharmaTrust)

Duration: Nov/Dec 2025
Role: Front-End Hardware Developer
Tools/Technologies: ESP32, DHT11, MQ-2 Gas Sensor, Pressure Sensor, IR Sensor, ThingSpeak Cloud, HTML, CSS, JavaScript, Embedded C/C++

SKILLS GAINED



PROJECTS

Real-Time Teleoperation Platform for Robotic Manipulator

Duration: Nov/Dec 2025
Role: Embedded Systems & Hardware Developer
Tools/Technologies: ESP8266, STM32 Nucleo Board, Servo Motors, PWM Driver Module, Embedded C/C++, HTML, CSS, JavaScript, Wi-Fi Networking

UI/UX for Accent Conversion of Speaker in Audio

Duration: July 2024 – November 2024
Role: UI/UX Designer and Developer
Tools/Technologies: Python, Pydub, MP3/WAV, Audio Processing

IoT-Based Attendance Monitoring System (ESP32-S3)

Duration: July 2024 – November 2024
Role: System Developer and Designer
Tools/Technologies: ESP32-S3 LilyGO, R307S Fingerprint Module, ThingSpeak, MATLAB

Advanced Glucose Monitoring System (Python, PyQt6)

Duration: December 2024 - April 2025
Role: Lead Developer & UI Designer
Tools/Technologies: Python, PyQt6, Matplotlib, NumPy, ThingSpeak (for cloud visualization)

Autonomous Robotic Arm for Industrial Material Handling

Duration: December 2024 - April 2025
Role: Embedded Systems Developer
Tools/Technologies: STM32 Nucleo-F103RB, SG90 Servo Motors, ESP8266 Wi-Fi Module, Potentiometers (B10-K), 3-DOF Robotic Arm, Power Supply Units (5V & 3.3V)

Sanitary Napkin Disposal Machine Aligned with SDG Goals

Duration: December 2023 – April 2024
Role: Project Designer and Tester
Tools/Technologies: Sensors, IoT Integration, Automated Control System, Embedded Systems

CERTIFICATIONS

- Completed Introduction to Linear Algebra with MATLAB on December 25, 2023:** This course covered fundamental concepts of linear algebra and its applications using MATLAB.
- Completed MATLAB Fundamentals on March 14, 2024:** Gained hands-on experience in MATLAB programming, data visualization, and basic scripting.
- Completed Solving Ordinary Differential Equations with MATLAB on March 21, 2024:** Learned to model, solve, and analyze ordinary differential equations using MATLAB.
- Completed MATLAB Programming Techniques on April 28, 2024:** Enhanced proficiency in MATLAB programming with advanced scripting and automation techniques.
- Completed Signal Processing Onramp on January 5, 2025:** Explored fundamental signal processing concepts and techniques using MATLAB.
- Completed Image Processing Onramp on February 9, 2025:** Gained practical experience in image manipulation, filtering, and enhancement in MATLAB.
- Completed Image Processing with MATLAB on March 5, 2025:** Studied advanced image processing techniques, feature extraction, and object recognition.

DECLARATION

I confirm that the above information is accurate and reflects my qualifications, experience, and skills to the best of my knowledge.

PUBLICATION & PATENT

“PyroSentinel: AI-Powered LoRa-Based Low Power Forest Fire Early Detection and Intelligent Communication System” - Patent Filed (March 2026). Designed and developed an ESP32 and LoRa-based intelligent forest fire detection system with multi-sensor fusion and AI-driven fire prediction.

“IntelliLift: Real-Time Occupancy Detection and Boarding Suggestions” - Presented at FOSGRIE 2026 Conference, Thiagarajar School of Management. Paper published in Scopus Indexed Conference Proceedings, focusing on smart elevator occupancy detection using IoT and AI-based vision systems.

SEMINARS & WORKSHOPS

- Printed Circuit Board Design & Analysis From an Idea to Product (Feb 26-28, 2025, Ramco Institute of Technology):** Attended a three-day national-level workshop at Ramco Institute of Technology, sponsored by TNSCST.
- Drone Technologies – Exploring the Skies (Sep 21-25, 2024, Kalasalingam Academy of Research and Education):** This program focused on drone technology fundamentals, flight mechanics, navigation, and applications in real-world scenarios.
- Remote Pilot Training Program (Aug 19, 2024, Kalasalingam Academy of Research and Education):** Conducted by Dhaksha Unmanned Systems (RPTO), a DGCA-approved organization, this session covered UAV regulations, flight operations.